

AGRIM SIGDEL

Full-Stack Developer | AI Researcher
Kathmandu, Nepal

sigdelagrim35@gmail.com | agrimsigdel.com.np | github.com/Agrim-Sigdel

PROFESSIONAL SUMMARY

Full-Stack Developer and published Computer Vision Researcher specializing in building end-to-end machine learning pipelines. Proven track record of designing advanced YOLO and RT-DETR architectures and deploying them into production-ready web platforms using Python/Django and React. Passionate about bridging the gap between theoretical AI research and scalable, user-centric cloud infrastructure.

TECHNICAL SKILLS

- **Frontend:** React.js, React Native, TypeScript, Next.js, Tailwind CSS, Framer Motion
- **Backend:** Python, Django, Django REST Framework (DRF), REST API Design
- **AI & CV:** Computer Vision, YOLO / RT-DETR, NLP / Transformers, PyTorch, BERT
- **Data & DevOps:** PostgreSQL / MySQL, Docker, CI/CD Pipelines, Git / GitHub / GitLab, Server Configuration, AWS (EC2, IAM), Grafana, Loki

RESEARCH & PUBLICATIONS

Cascaded AI Task Decomposition (CATD)

2024–2025

Lead Researcher & Developer

- **Publication:** Gajurel, R. K., Sigdel, A., Magar, B. T., & Thapa, S. (2026). Cascaded AI Task Decomposition: A Generalizable Framework for Multi-Stage Computer Vision Pipeline Validated on Automatic Number Plate Recognition for Nepali License Plates. *National College of Computer Studies Research Journal*, 5(1), 137-167.
- Achieved 97.1% mAP@50 plate localization and 97.8% mAP@50 character recognition using custom-trained YOLOv11n and RT-DETR Base on 61-class bi-scriptural Nepali plates.
- Built a two-dimensional cross-domain training strategy (1,380 Nepali + 1,127 Indian plates) enabling compositional zero-shot transfer (+24.5 pp improvement).
- Engineered a three-stage cascaded pipeline (YOLOv8n → YOLOv11n → RT-DETR) reaching 90.4% end-to-end precision with 61%-73% compute reduction via intelligent vehicle gating.
- Integrated the system as the computer-vision backbone of a smart-parking platform handling automated billing and real-time traffic analytics.

WORK EXPERIENCE

Contractor Ops AI

Jan 2026–Present

Software Developer (Full-Stack)

Remote

- Engineer a high-performance mobile application using React Native and TypeScript, developing and shipping diverse full-stack features across the entire application stack.
- Deploy and manage backend & AI services on AWS EC2, utilizing AWS IAM to securely manage roles and access permissions.
- Implement centralized system monitoring and observability by configuring log draining to Grafana via Loki.
- Debug and resolve full-stack integration bottlenecks, optimizing database query handling, efficient serialization, and low-latency payloads between server and client.

Kingsoft Tech

Oct 2025–Jul 2026

Full-Stack Developer

Kathmandu, Nepal (On-site)

- Spearheaded full-stack development, optimization, and deployment for Feriaau and its multi-tenant CRM platform.
- Owned the complete lifecycle of modular Python/Django backend services, robust REST APIs, and dynamic React.js/TypeScript interfaces.
- Configured secure production servers and automated CI/CD pipelines, ensuring high availability and seamless code delivery.
- Optimized PostgreSQL/MySQL schemas and queries to maximize application speed and multi-tenant isolation.

Self-Employed

2023–Present

Freelance Developer

Remote

- Designed, built, and shipped a range of client projects end-to-end, including responsive marketing websites and comprehensive full-stack web applications for diverse clients.
- Developed bespoke CRMs and tailored internal business management tools utilizing React, TypeScript, and Python/Django to streamline client workflows and automate operational processes.
- Architected scalable REST APIs and optimized database schemas to support multi-tenant data structures and real-time features for custom freelance applications.
- Handled the full delivery pipeline including frontend development, secure backend infrastructure, cloud deployment, and post-launch maintenance across varied tech stacks.

Prime College Graduation Committee

May 2024–Sep 2024

Creative Lead

Kathmandu, Nepal

- Led creative direction and brand guidelines for 625+ attendees, coordinating event planning and execution.
- Managed cross-functional teams for visual and marketing materials, practicing agile task delegation.

PROJECTS & OPEN SOURCE

Partweave – Open Source CLI Tooling

Creator

- Creator of an open-source full-stack TypeScript CLI that generates self-contained monorepos with deterministic template injection (Django, Next.js, Expo, Auth, Docker).

easy-3dkit – NPM Package

Creator & Maintainer

- Creator and maintainer of an open-source library providing interactive 3D components for React, built natively on Three.js, React Three Fiber (R3F), and GSAP.

Reddit Sentiment Analysis (NLP)

Developer

- Deep NLP pipeline using fine-tuned BERT for nuanced emotion classification on large-scale Reddit data, integrated with LLM-based frontend interfaces.

EDUCATION

Prime College, Tribhuvan University

2022–2026

BSc. Computer Science and Information Technology (CSIT)

Kathmandu, Nepal

- **Aggregate Score:** 75.3%